

5.3 Operating conditions

5.3.1 General operating conditions

Table 24. General operating conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V _{DD}	Standard operating voltage	-	2.0 ⁽¹⁾	3.6	V
V _{DDIO2}	External supply voltage for selected I/Os	-	1.65	3.6	V
V _{IN}	I/O input voltage	-	-0.3	Min (V _{DDIOx} + 3.6, 5.5) ⁽²⁾⁽³⁾	V
f _{HCLK}	AHB clock frequency	-	-	48	MHz
f _{PCLK}	APB clock frequency	-	-	48	MHz
T _A	Ambient temperature ⁽⁴⁾	Suffix 6 ⁽⁵⁾	-40	85	°C
		Suffix 7 ⁽⁵⁾	-40	105	
		Suffix 3 ⁽⁵⁾	-40	125	
T _J	Junction temperature	Suffix 6 ⁽⁵⁾	-40	105	°C
		Suffix 7 ⁽⁵⁾	-40	125	
		Suffix 3 ⁽⁵⁾	-40	130	

1. When RESET is released, functionality is guaranteed down to V_{PDR}.
2. For operation with voltage higher than V_{DD} + 0.3 V, the internal pull-up and pull-down resistors must be disabled.
3. When V_{DDIO2} > V_{DD} and an FT_sa pin is used by an analog peripheral such as ADC, the maximum V_{IN} is V_{DD} + 0.3V.
4. The T_A(max) applies to P_D(max). At P_D < P_D(max) the ambient temperature is allowed to go higher than T_A(max) provided that the junction temperature T_J does not exceed T_J(max). Refer to [Section 6.11: Thermal characteristics](#).
5. Temperature range digit in the order code. See [Section 7: Ordering information](#).

5.3.2 Operating conditions at power-up / power-down

The parameters given in [Table 25](#) are derived from tests performed under the ambient temperature condition summarized in [Table 24](#).

Table 25. Operating conditions at power-up / power-down

Symbol	Parameter	Min	Max	Unit
t _{VDD}	V _{DD} rise time rate	0	∞	μs/V
	V _{DD} fall time rate	10	∞	

5.3.3 Embedded reset and power control block characteristics

The parameters given in [Table 26](#) are derived from tests performed under the ambient temperature conditions summarized in [Table 24](#).